4 *Managing conversation and state in your application 88*

4.1 AI SDK React server components 90

Overview of RSCs 90 ▪ Using server actions for AI-powered RSCs 92 ▪ Updating the UI to use server actions 96
Techniques for generating and streaming UI components 98
Creating streamable UI components from LLM providers with streamUI 99 ▪ Streaming React components with createStreamableUI 101

4.2 Managing UI state in AI-powered applications 103

Separating AI and UI state in React/Next.js applications 103
Key components for UI state management 105
Implementing UI state management patterns 108

4.3 Structured data generation using the Vercel AI SDK 113

How structured data generation works 113 ▪ Techniques for generating structured data from AI responses 114
Tools for implementing type-safe AI-generated content 115
Integrating structured data generation into our web application 117

4.4 Tool and function calling with AI models 121

Understanding tool calling and function calling in AI models 121
Implementing custom tools and functions with the Vercel AI SDK 123

PART 2 ADVANCED GENERATIVE AI TECHNIQUES AND DEPLOYMENT 127

5 *Prompt engineering in web applications 129*

5.1 Introducing prompt engineering 131

What exactly are prompts? 132 ▪ Prompt types 137
Organizing your prompts: Versioning, testing, and optimization 139

5.2 Few-shot learning 146

Examples of few-shot learning 147 ▪ General methodology for creating few-shot learning prompts 150

5.3 Chain-of-thought prompting: A deeper dive into reasoning 151

Example of chain-of-thought prompting 151 ▪ General methodology for creating chain-of-thought prompts 152